

BFS

Initial State :

1	2	3
	4	6
7	5	8

Goal State :

1	2	3
4	5	6
7	8	

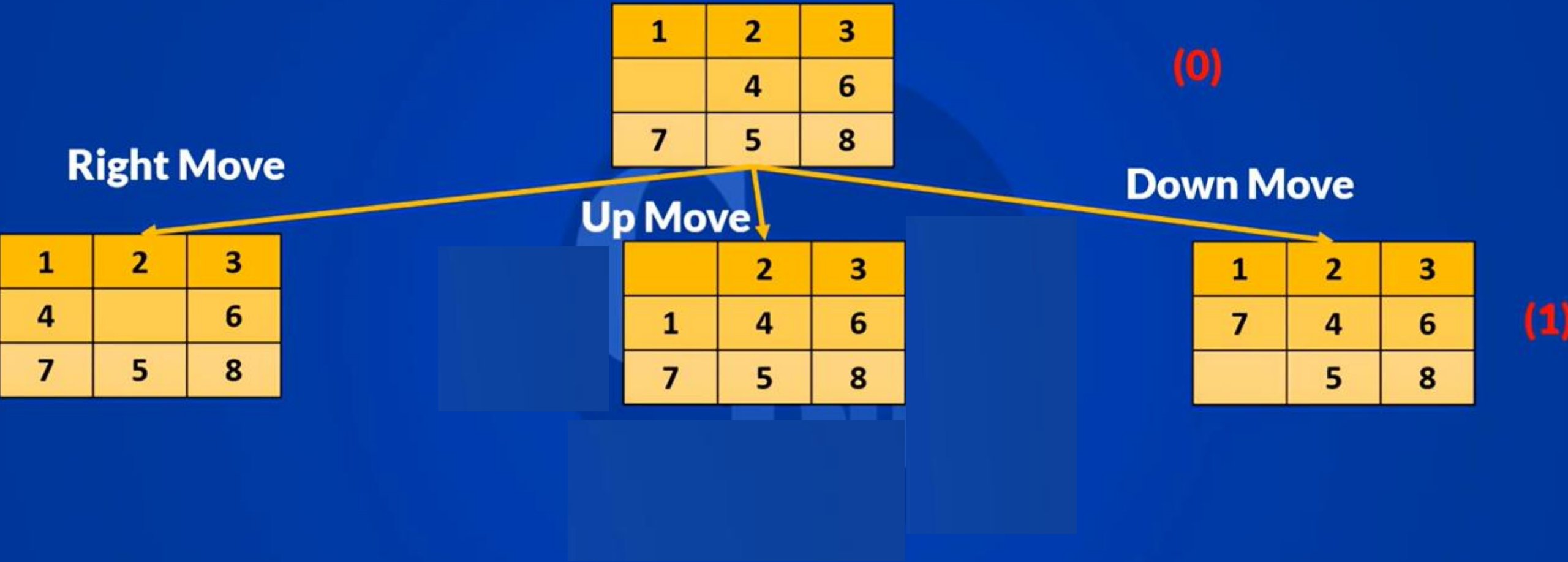
- Uninformed Search ,Non-Heuristic Search (BFS or DFS)
- Complexity : $O(b^d)$, where
b=branching factor
d=depth factor
- Worst case: 3^{20}
- For 8 puzzle Problem Branching Factor:
(All possible moves of empty tile at each position)/(No.of Tiles)

Breath First Search to solve Eight puzzle problem:

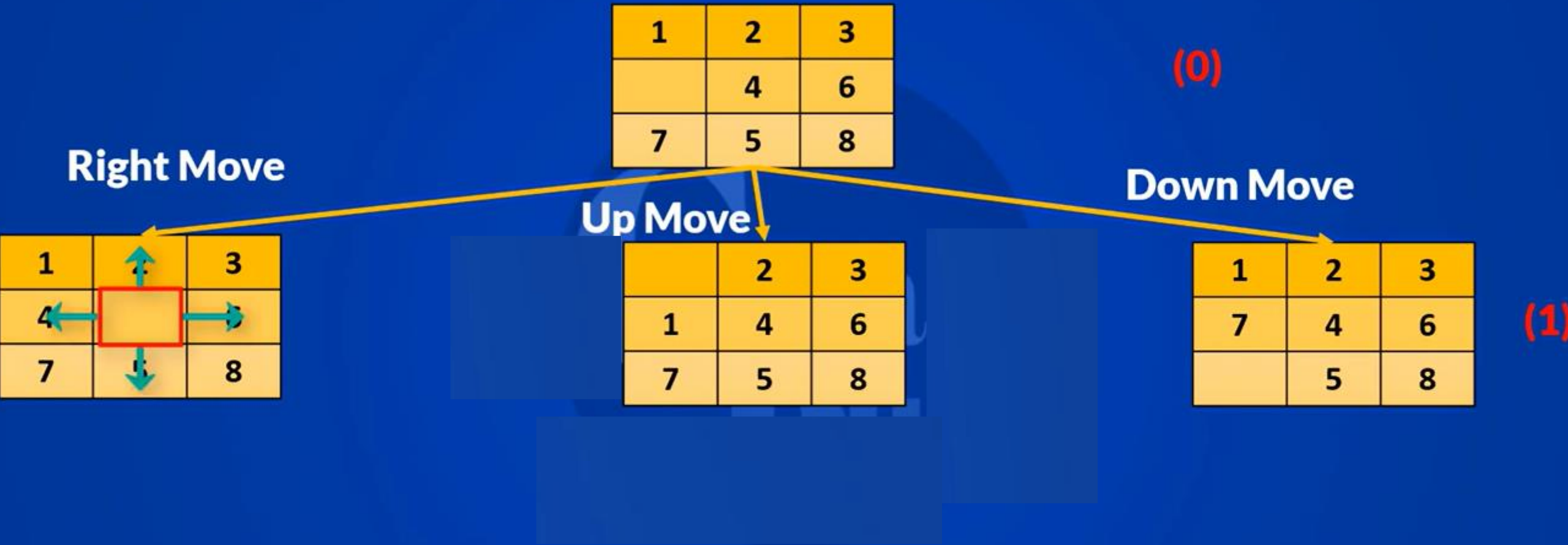
1	2	3
	4	6
7	5	8

(0)

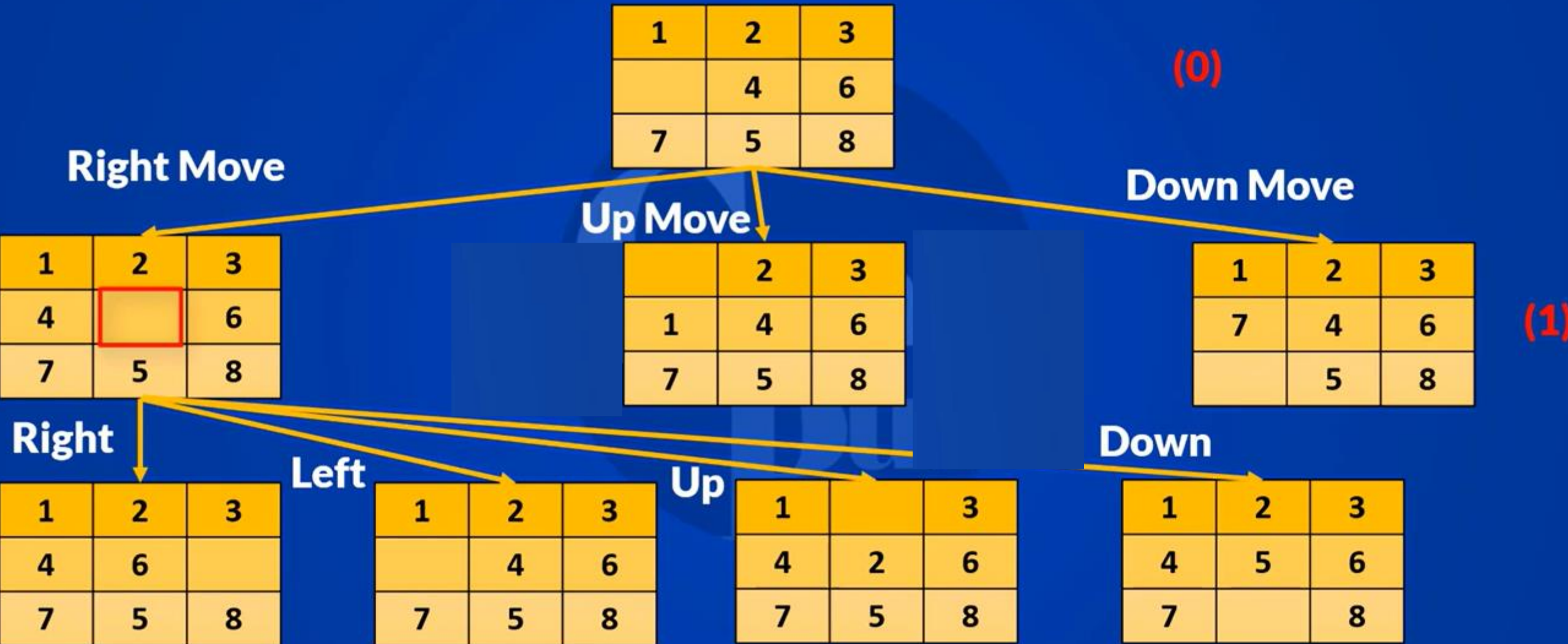
Breath First Search to solve Eight puzzle problem:



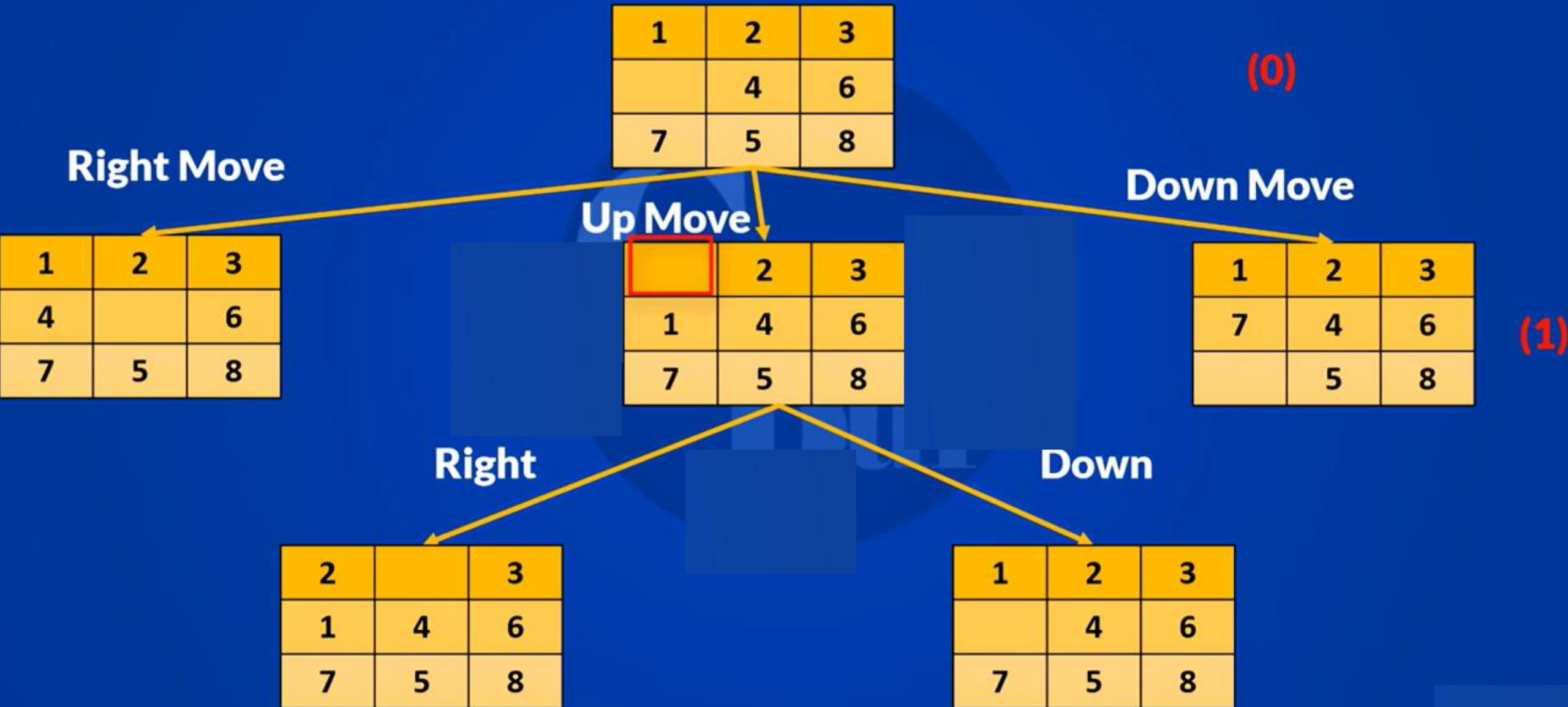
Breath First Search to solve Eight puzzle problem:



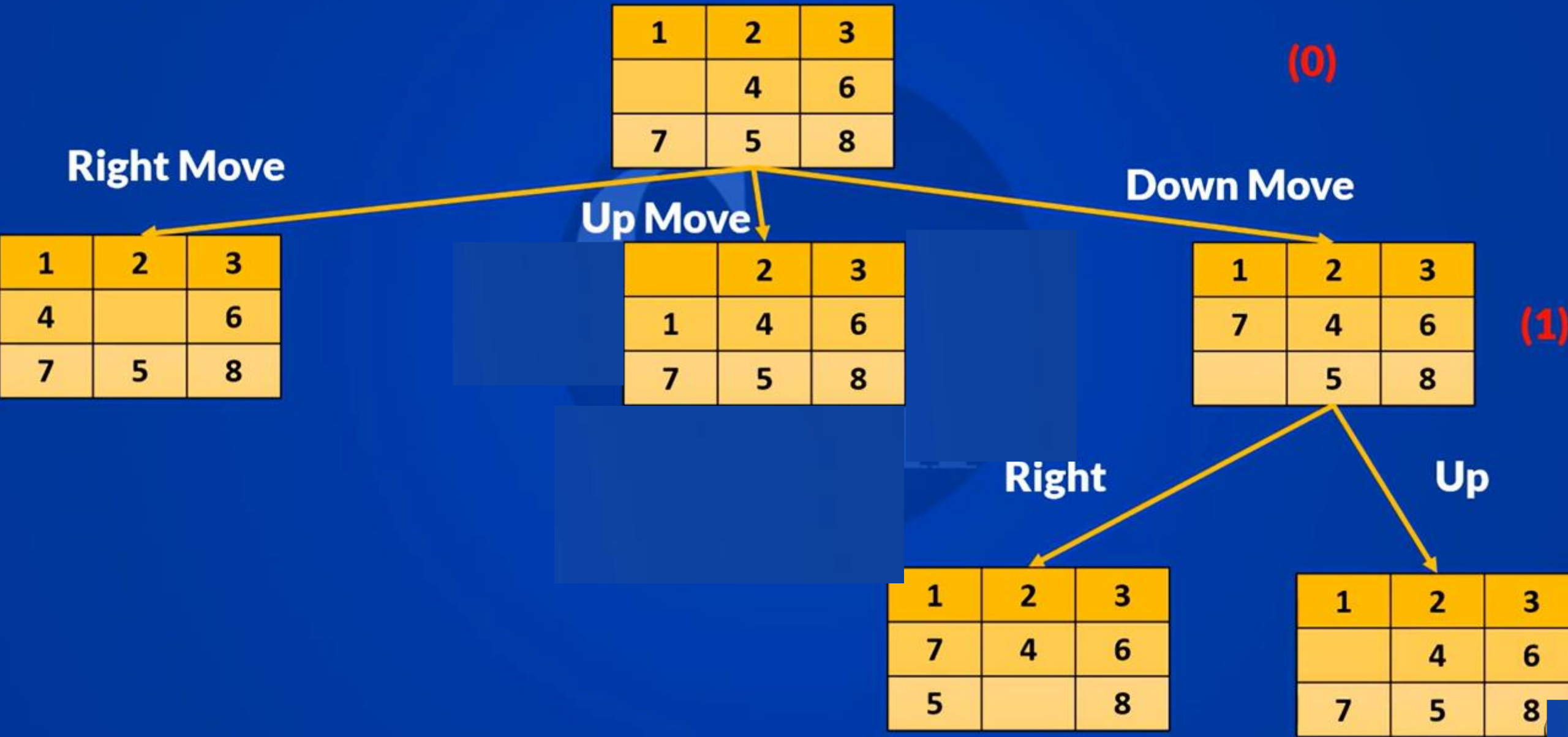
Breath First Search to solve Eight puzzle problem:



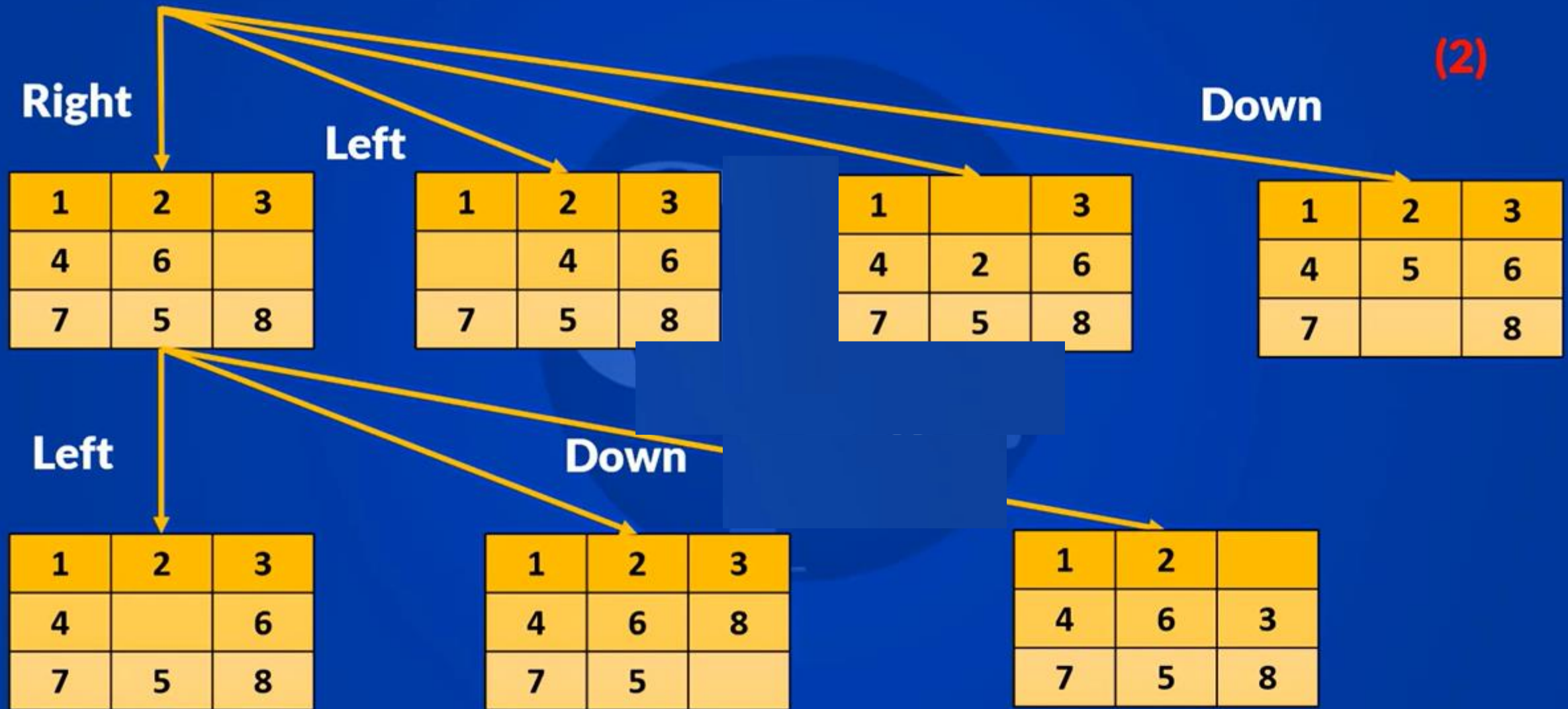
Breath First Search to solve Eight puzzle problem:



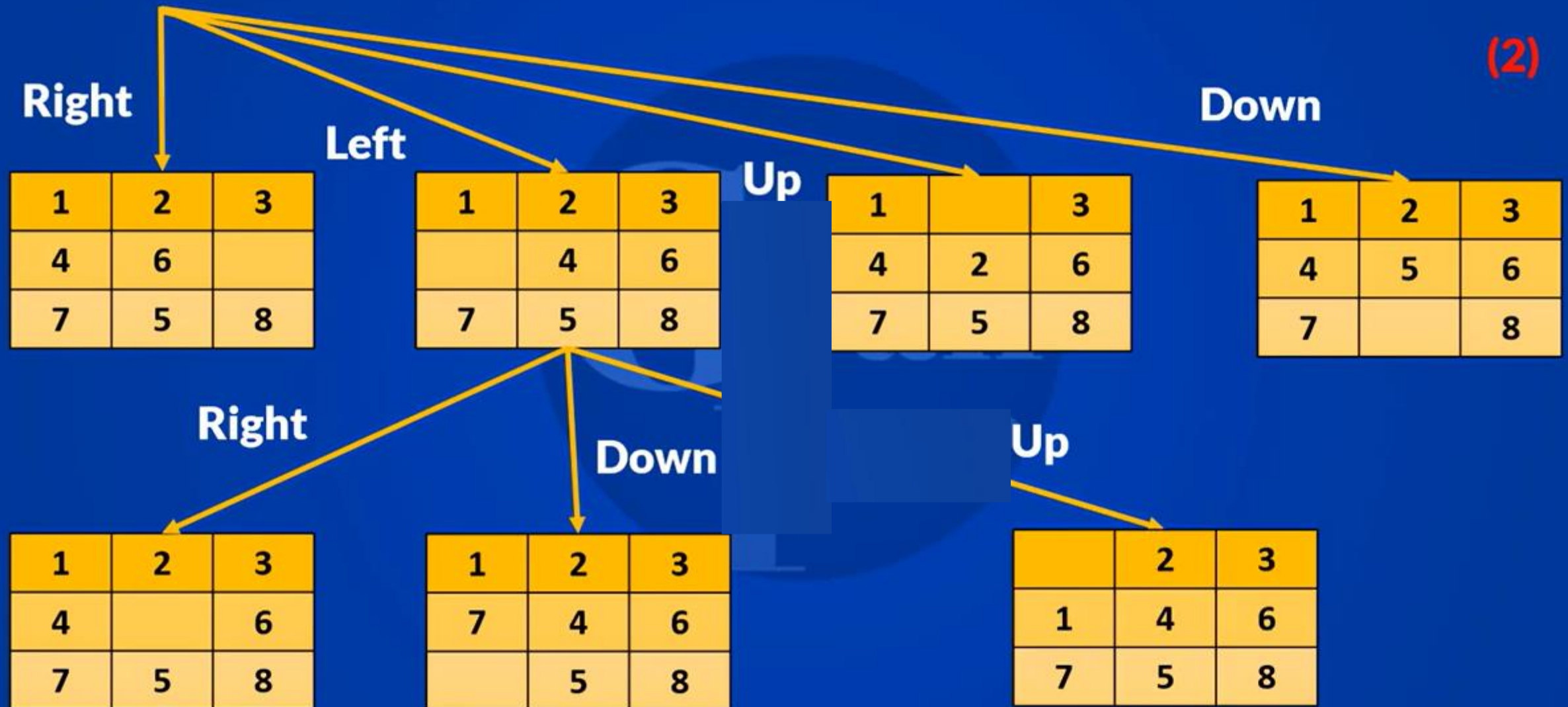
Breath First Search to solve Eight puzzle problem:



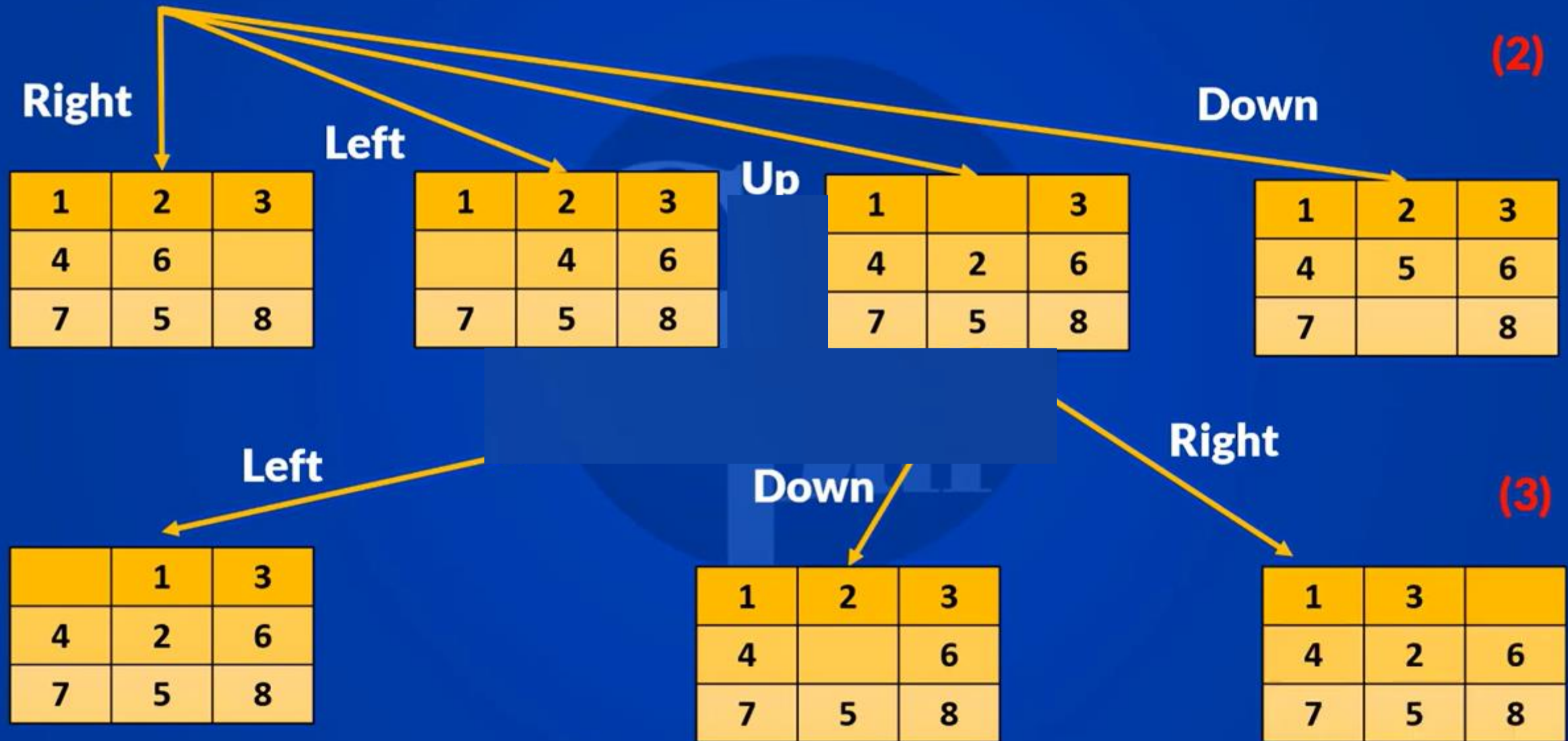
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